Finite Trigonometry over $\mathbb{F}_7$

Angles in $\mathbb{Z}/8$, the Conic $x^2 + y^2 = 1$, and a First Look at Harmonic Analysis

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Abstract

Over $\mathbb{F}_7$ the conic $x^2 + y^2 = 1$ has exactly eight points. They form the norm-one group of $\mathbb{F}_{49} = \mathbb{F}_7[\mathrm{i}]/(\mathrm{i}^2 + 1)$, a cyclic group of order eight. Writing $\alpha = 2 + 2\mathrm{i}$ and $\alpha^k = \cos k + \mathrm{i} \sin k$ defines a finite trigonometry with angle group $\mathbb{Z}/8$ and values in $\mathbb{F}_7$. We prove the Pythagorean, addition, doubling and periodicity identities, draw the finite unit circle and the angle group, give the stereographic (half-angle) coordinate, and draw the discrete waveforms, and compare the finite angle group with the Euclidean angle axioms. A final section introduces the harmonic layer: the characters of $\mathbb{Z}/8$, the finite Fourier transform, and the difference between the $\mathbb{F}_7$-valued trigonometric functions and the $\mathbb{C}$-valued characters. We also explain why the angle measure is discrete and why half-angles need not exist. This is an algebraic and finite theory; it is not a limit-based real analysis, and the note says explicitly what is and is not claimed.

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1 Introduction

Over the real numbers the functions $\cos\theta$ and $\sin\theta$ are defined by the unit circle $x^2 + y^2 = 1$ and by angles in $\mathbb{R}/2\pi\mathbb{Z}$. Over a finite field the circle is a finite set, the angle group is finite, and the values are field elements. The algebraic identities survive; the limiting and continuity notions do not.

For $\mathbb{F}_7$ the situation is particularly clean. The conic

$$C: \quad x^2 + y^2 = 1$$

has exactly $7+1 = 8$ points, and these points form a cyclic group of order 8 under the multiplication of the field $\mathbb{F}_{49}$. This gives a finite analogue of the circle group, with angle group $\mathbb{Z}/8$, and a finite trigonometry whose identities are polynomial identities over $\mathbb{F}_7$. Finite-field trigonometry has been studied in [4]; the present note is a concrete geometric model, with explicit figures and computational checks.

The plan is:

1. define the field $\mathbb{F}_{49}$ and the norm-one group, and ask whether the norm induces a metric or a topology (Section 2);
2. define $\cos k$ and $\sin k$, and prove the basic identities (Sections 3–4);
3. draw the finite unit circle and the angle group (Sections 5–6);
4. compare the finite angle group with the Euclidean angle axioms (Section 7);
5. give the stereographic/half-angle coordinate (Section 8);
6. draw the discrete waveforms (Section 9);
7. introduce the harmonic layers: finite-field characters in $\mathbb{F}_{49}$ and the complex Fourier transform (Section 10);
8. state what is not claimed and what comes next (Section 11).

Throughout, $\mathbb{F}_7 = \{0, 1, \ldots, 6\}$ with arithmetic modulo 7; standard background on finite fields is in [3]. We refer to [6] and [7] as Paper 0 and Paper I. The element i is a formal square root of -1; since -1 is not a square in $\mathbb{F}_7$ (the squares are $0, 1, 2, 4$), the ring $\mathbb{F}_7[\mathrm{i}]/(\mathrm{i}^2 + 1)$ is a field with 49 elements.

2 The field $\mathbb{F}_{49}$ and the norm-one group

Definition 2.1 (Norm and norm-one group)**.** For $z = x + y\mathrm{i} \in \mathbb{F}_{49}$ define the norm

$$N(z) = z\bar{z} = (x + y\mathrm{i})(x - y\mathrm{i}) = x^2 + y^2 \in \mathbb{F}_7.$$

Let $G = \{z \in \mathbb{F}_{49}^\times : N(z) = 1\}$.

Proposition 2.2. *G is a cyclic group of order 8.*

Proof. The norm map $N : \mathbb{F}_{49}^\times \to \mathbb{F}_7^\times$ is a surjective homomorphism of finite groups (the norm of a generator of $\mathbb{F}_{49}^\times$ generates $\mathbb{F}_7^\times$). Hence $|G| = 48/6 = 8$. A subgroup of the cyclic group $\mathbb{F}_{49}^\times$ is cyclic, so G is cyclic. □

Proposition 2.3. *Let $\alpha = 2 + 2\mathrm{i}$. Then α has order 8, and $G = \langle\alpha\rangle$.*

Proof. We compute

$$\alpha^2 = 4 + 8\mathrm{i} + 4\mathrm{i}^2 = 8\mathrm{i} = \mathrm{i}, \qquad \alpha^4 = \mathrm{i}^2 = -1, \qquad \alpha^8 = 1.$$

Since $\alpha^2 = \mathrm{i} \neq 1$ and $\alpha^4 = -1 \neq 1$, the order of α is 8. It lies in G because $N(\alpha) = 2^2 + 2^2 = 8 = 1$, and G has order 8; therefore $G = \langle\alpha\rangle$. □

The eight powers are

k	0	1	2	3	4	5	6	7
α^k	1	$2 + 2\mathrm{i}$	i	$5 + 2\mathrm{i}$	-1	$-2 - 2\mathrm{i}$	$-\mathrm{i}$	$2 - 2\mathrm{i}$

and their coordinates (x, y) are the eight points

$$(1, 0),\ (2, 2),\ (0, 1),\ (5, 2),\ (6, 0),\ (5, 5),\ (0, 6),\ (2, 5).$$

They are exactly the solutions of $x^2 + y^2 = 1$ in $\mathbb{F}_7^2$.

2.1 Does the norm induce a metric or a topology?

The norm takes values in $\mathbb{F}_7$, not in an ordered set. A metric in the analytic sense is a function d with values in an ordered set, positive on distinct points and satisfying the triangle inequality. The candidate $d(z, w) = N(z - w)$ fails at the first step: $\mathbb{F}_7$ has no canonical order, so "$N(z - w) \geq 0$" and "$N(z - w) < r$" are not meaningful. The norm is anisotropic ($N(z) = 0$ iff $z = 0$, because -1 is a nonsquare), but anisotropy is nondegeneracy, not a metric.

The only absolute value on $\mathbb{F}_7$ is the trivial one ($|0| = 0$, $|x| = 1$ for $x \neq 0$). Composing it with the norm gives $|N(z - w)| \in \{0, 1\}$, the discrete metric; the induced topology is the discrete topology. The Zariski topology on the finite set of $\mathbb{F}_7$-points is also discrete. In this analytic sense the norm gives no interesting metric or topology.

In the algebraic sense the answer is different and richer. The norm is a nondegenerate anisotropic quadratic form; its polar form is the dot product $u \cdot v$, and its isometry group is the orthogonal

group $\mathrm{O}_2(\mathbb{F}_7)$. The rotation subgroup is exactly the norm-one group $G = \mathrm{SO}_2(\mathbb{F}_7) \cong \mathbb{Z}/8$, and the conic C is the unit sphere. So the norm does induce a *metric vector space* (a quadratic space) with orthogonality, isometries and a unit sphere; it does not induce a metric space in the topological sense.

What the norm does induce geometrically is a stratification. The fibres are

$$N^{-1}(0) = \{0\}, \qquad N^{-1}(a) \cong C \quad (a \in \mathbb{F}_7^\times),$$

each nonzero fibre being an 8-point conic and a G-torsor under $z \mapsto \alpha^k z$. The six nonzero fibres partition $\mathbb{F}_{49}^\times$; Figure 1 shows them. This is the finite analogue of polar coordinates: radius $a \in \mathbb{F}_7^\times$, angle $k \in \mathbb{Z}/8$. The decomposition is a torsor decomposition, not a canonical group product: the exact sequence $1 \to \mathbb{Z}/8 \to \mathbb{F}_{49}^\times \to \mathbb{F}_7^\times \to 1$ does not split as a direct product (the two orders have gcd 2), and choosing a base point in each fibre is a noncanonical section.

The cyclic word metric on $G = \mathbb{Z}/8$ (the one used for the angle group) is not induced by the norm: the norm is constant 1 on G. It is induced by the cyclic order of the angle group.

To obtain a genuine metric and topology from the norm one must lift. The unramified extension $\mathbb{Q}_{49}/\mathbb{Q}_7$ has residue field $\mathbb{F}_{49}$; the 7-adic absolute value gives a metric on $\mathbb{Q}_{49}$, the field norm $N_{\mathbb{Q}_{49}/\mathbb{Q}_7}$ is continuous, and the finite norm is its reduction modulo 7. This is the natural bridge from the finite norm to p-adic analysis; the finite field itself only carries the discrete topology.

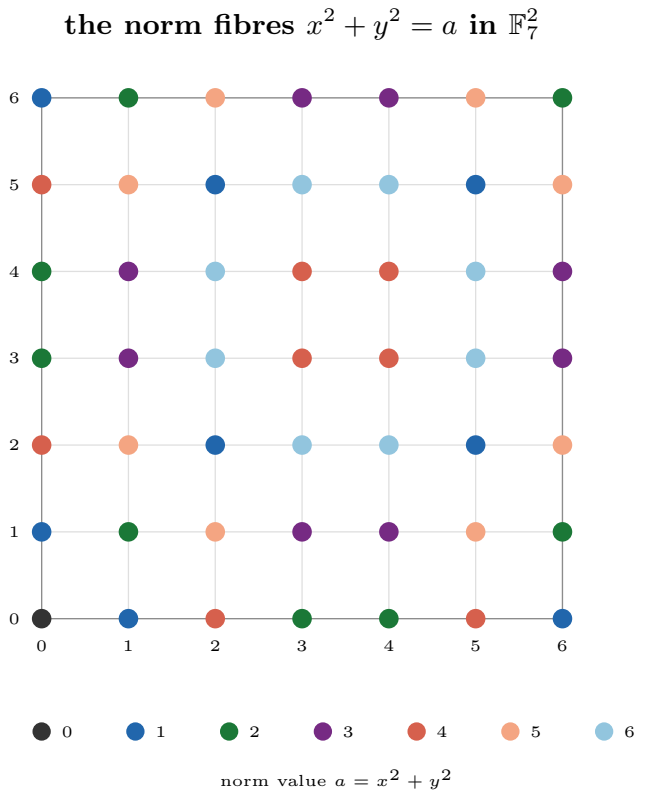


Figure 1: The norm fibres $x^2+y^2 = a$ in $\mathbb{F}_7^2$: the origin ($a = 0$) and six 8-point conics ($a = 1, \ldots, 6$). The colours are the norm value; the legend is at the bottom. This is a stratification, not a metric topology.

question	answer
distance $d(z,w) = N(z-w)$ in an ordered set?	no: $\mathbb{F}_7$ is unordered; only the trivial absolute value; discrete topology
metric vector space?	yes: nondegenerate anisotropic quadratic form; dot product; $\mathrm{O}_2(\mathbb{F}_7)$
topology from the norm?	discrete; the Zariski topology on the finite points is discrete
fibres?	$\{0\}$ and six 8-point conics; G-torsors
genuine metric or topology?	lift to $\mathbb{Q}_{49}/\mathbb{Q}_7$ or $W(\mathbb{F}_{49})$; 7-adic absolute value
metric on $\mathbb{Z}/8$?	cyclic word metric, from the group, not from N

Table 1: What the norm does and does not induce. The analytic notions (distance, order, continuity) are absent; the algebraic notions (quadratic form, isometries, torsors) are present; the analytic version requires a lift.

3 Cos and sin

Definition 3.1 (Finite cosine and sine)**.** For $k \in \mathbb{Z}/8$, write

$$\alpha^k = \cos k + \mathrm{i} \sin k, \qquad \cos k,\ \sin k \in \mathbb{F}_7.$$

The values are given in Table 2.

k	α^k	$\cos k$	$\sin k$
0	1	1	0
1	$2+2\mathrm{i}$	2	2
2	i	0	1
3	$5+2\mathrm{i}$	5	2
4	-1	6	0
5	$-2-2\mathrm{i}$	5	5
6	$-\mathrm{i}$	0	6
7	$2-2\mathrm{i}$	2	5

Table 2: The eight angles and their cosines and sines in $\mathbb{F}_7$.

Proposition 3.2 (Basic identities)**.** *For all $k, l \in \mathbb{Z}/8$:*

1. *(periodicity)* $\cos(k+8) = \cos k$ *and* $\sin(k+8) = \sin k$*;*
2. *(parity)* $\cos(8-k) = \cos k$ *and* $\sin(8-k) = -\sin k$*;*
3. *(Pythagoras)* $\cos^2 k + \sin^2 k = 1$.

Proof. (1) is $\alpha^{k+8} = \alpha^k$. (2) is $\alpha^{-k} = \overline{\alpha^k}$, using $\bar{\alpha} = \alpha^{-1}$. (3) is $N(\alpha^k) = 1$. □

Remark 3.3. The values are in $\mathbb{F}_7$ and the angle is in $\mathbb{Z}/8$; there is no continuous graph and no real-valued sine wave. The identities are polynomial identities in $\mathbb{F}_7$, and they are verified for all k by the script `verify.py`.

3.1 The angle measure is discrete; there are no radians

The protractor is the isomorphism

$$\mathbb{Z}/8 \longrightarrow G, \qquad k \longmapsto \alpha^k.$$

The angle is the group element; the label k is its coordinate. This is the whole of the finite angle measure. There is no real-valued additive measure, no arc length, and no radian: $\mathbb{F}_7$ is unordered, the conic is a finite set, and there is no continuous arc to length.

One can put a metric on $\mathbb{Z}/8$, the cyclic word metric

$$\ell(k) = \min(k, 8-k),$$

but it is a size, not a measure: it is not additive. For example, $\ell(3+4) = \ell(7) = 1$ while $\ell(3) + \ell(4) = 3 + 4 = 7$. The only additive "measure" is the group element itself; the word metric is the metric of the Cayley graph, not a protractor.

The unit is the choice of generator: α generates G, and the automorphisms of $\mathbb{Z}/8$ are multiplication by $1, 3, 5, 7$. Changing the generator relabels the eight angles but does not change the group. The full turn is $8 = 0$, the straight angle is 4, and the right angle is 2; calling the generator a "degree" is a labelling choice, not a radian. The periodic waveforms of Figure 11 show the period 8; there is no interpolation between the eight points.

4 Addition and doubling

Proposition 4.1 (Addition formulas)**.** *For all $k, l \in \mathbb{Z}/8$,*

$$\cos(k+l) = \cos k \cos l - \sin k \sin l, \qquad \sin(k+l) = \sin k \cos l + \cos k \sin l.$$

Proof. Multiply:

$$\alpha^{k+l} = (\cos k + \mathrm{i} \sin k)(\cos l + \mathrm{i} \sin l) = (\cos k \cos l - \sin k \sin l) + \mathrm{i}(\sin k \cos l + \cos k \sin l).$$

The real and imaginary parts are the two displayed identities. □

Corollary 4.2 (Doubling)**.** *For all $k \in \mathbb{Z}/8$,*

$$\cos 2k = \cos^2 k - \sin^2 k, \qquad \sin 2k = 2 \sin k \cos k.$$

The addition formula is the group law of $\mathbb{Z}/8$ transported to the conic: the map

$$k \longmapsto (\cos k, \sin k) = \alpha^k$$

is an isomorphism $\mathbb{Z}/8 \to G = C(\mathbb{F}_7)$. The doubling map $k \mapsto 2k$ is an endomorphism of $\mathbb{Z}/8$ with kernel $\{0, 4\}$; it is two-to-one, just like the real doubling map on the circle. It is *not* an automorphism (the automorphisms of $\mathbb{Z}/8$ are multiplication by $1, 3, 5, 7$).

Example 4.3. For $k = 1$: $\cos 1 = 2$, $\sin 1 = 2$, so $\cos 2 = 4 - 4 = 0$ and $\sin 2 = 2 \cdot 2 \cdot 2 = 8 = 1$, which is the table entry for $k = 2$.

4.1 Half-angles and division by two

The doubling formula gives a map $\mathbb{Z}/8 \to \mathbb{Z}/8$, $k \mapsto 2k$. Its kernel is $\{0, 4\}$ and it is two-to-one (Figure 8). The half-angle equation is

$$2h = k.$$

It is solvable if and only if k is even; when solvable it has two solutions, h and $h + 4$. For example, the right angle $k = 2$ has $h = 1, 5$, the straight angle $k = 4$ has $h = 2, 6$, and the odd angles $k = 1, 3, 5, 7$ have no half-angle at all.

The point is the difference between the field and the angle group:

- in the field $\mathbb{F}_7$, 2 is invertible ($2^{-1} = 4$), so $2x = a$ has a unique solution for every a;
- in the angle group $\mathbb{Z}/8$, 2 is not invertible, so $2h = k$ may have no solution, and when it has one it has two.

This is why the half-angle formula for the *square* of the cosine is well-defined even when the half-angle is not:

$$\cos^2 h = \frac{1 + \cos k}{2} \qquad (2h = k),$$

because the right-hand side only uses the field inverse of 2. The two solutions h and $h + 4$ give $(\cos h, \sin h)$ and $(-\cos h, -\sin h)$; the half-angle is defined only up to sign.

Table 3 summarizes the two equations. The consequences are:

- there is no global half-angle function $\mathbb{Z}/8 \to \mathbb{Z}/8$;
- there is no global angle bisector (the bisector exists only for even k, and then it is a line with two orientations);
- the inscribed-angle theorem has only a two-to-one shadow (Section 7);
- the stereographic coordinate t (the slope from P_0, the finite analogue of $\tan(k/2)$) is a rational coordinate on the seven angles $k \neq 4$, not a group homomorphism.

In Euclidean geometry the angle group $\mathbb{R}/2\pi\mathbb{Z}$ is divisible: $2h = k$ always has two solutions, h and $h + \pi$. The finite group $\mathbb{Z}/8$ is not two-divisible, and the failure is not a technicality: it is the absence of division in $\mathbb{Z}/8$.

equation	in the field $\mathbb{F}_7$	in the angle group $\mathbb{Z}/8$
$2x = a$	unique solution $x = a/2$	2 if a even; 0 if a odd
half-angle $2h = k$	unique $h = k/2$	2 if k even; 0 if k odd

Table 3: The field inverse of 2 and the angle equation $2h = k$. The field has $2^{-1} = 4$; the group $\mathbb{Z}/8$ has no division by 2.

5 The finite unit circle

The conic C is isomorphic to the projective line $\mathbb{P}^1(\mathbb{F}_7)$; the eight points of $C(\mathbb{F}_7)$ are the eight points of $\mathbb{P}^1(\mathbb{F}_7)$. The angle k is one coordinate on this set; the stereographic coordinate t of Section 8 is another. Neither coordinate is more intrinsic than the other: the intrinsic object is the conic together with its group law.

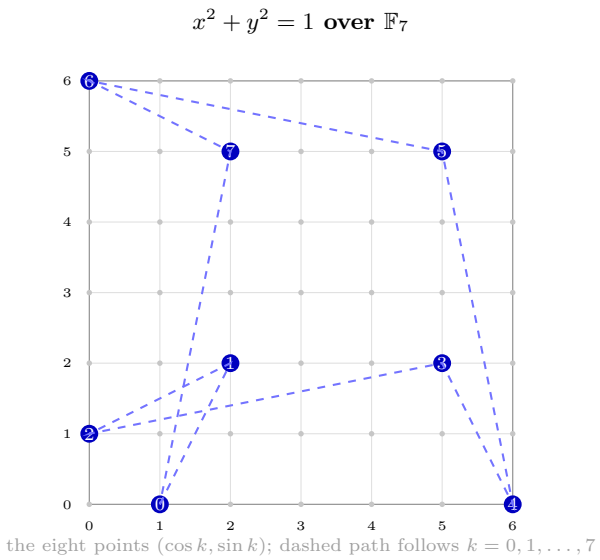


Figure 2: The eight points of $x^2 + y^2 = 1$ over $\mathbb{F}_7$, labelled by the angle $k \in \mathbb{Z}/8$. The dashed path follows $k = 0, 1, \ldots, 7$. This is a conic, not a Euclidean circle: the eight points are scattered in the 7×7 grid.

5.1 Centre, angle group, and order

Does a centre exist? Homogenize the conic as $C : x^2 + y^2 = z^2$ and let $L_\infty : z = 0$. The polar of $[a : b : c]$ is $ax + by - cz = 0$, so the pole of L_∞ is $O = [0 : 0 : 1] = (0, 0)$. The point O is not on C ($0 \neq 1$); it is the unique point with $N(O) = 0$, while every point of C has norm 1. The central symmetry $\sigma(x, y) = (-x, -y)$ preserves C and fixes O; on the angle group it is α^4, the half-turn, and it pairs the eight points into four antipodal pairs $\{k, k + 4\}$ (Figure 3).

The centre is finite exactly when L_∞ is not tangent to C: the ellipse and the hyperbola have a finite centre, while the parabola's centre is the tangency point at infinity.

The angle group around the centre. The symmetries of C fixing O are the orthogonal group $\mathrm{O}_2(\mathbb{F}_7)$ of the quadratic form $x^2 + y^2$, a dihedral group of order 16. Its rotation subgroup is

$$G = \mathrm{SO}_2(\mathbb{F}_7) \cong \mathbb{Z}/8,$$

the norm-one group; it acts regularly on C. The angle of a point P relative to a base point P_0 is the unique $k \in \mathbb{Z}/8$ with $P = \alpha^k P_0$; the rotations are the angle group around the centre.

The full conic automorphism group is $\mathrm{PGL}_2(7)$ of order 336. It does not fix O: the stabilizer of O is $\mathrm{O}_2(\mathbb{F}_7)$ of order 16, so the orbit of O has $336/16 = 21$ points, the internal points of the polarity. Thus the centre is extra structure: the conic alone is $\mathbb{P}^1$ with $\mathrm{PGL}_2(7)$; a centre (equivalently a line at infinity) selects the angle group. This is the same point as the earlier frame discussion: the "circle" is not just the conic.

Order: cyclic, not total. The angle group $\mathbb{Z}/8$ is cyclic; a generator α gives a cyclic order

$$0, 1, 2, \ldots, 7$$

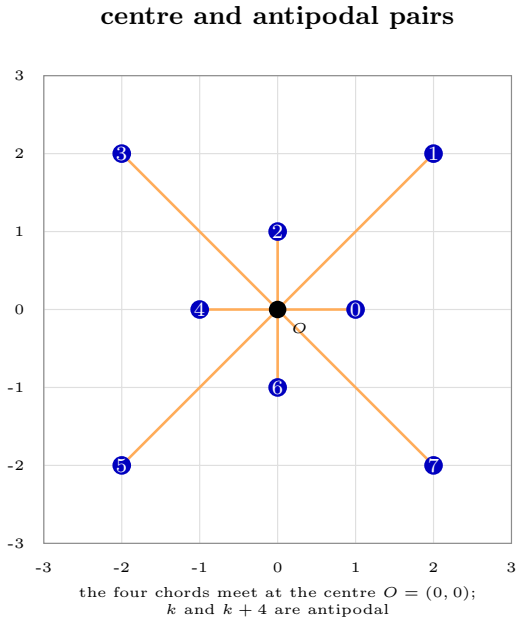


Figure 3: The centre $O = (0, 0)$ is the pole of L_∞. The four chords through O join the antipodal pairs k and $k + 4$. The centre is not on the conic.

on C. The rotations preserve this cyclic order, the reflections in $\mathrm{O}_2(\mathbb{F}_7)$ reverse it, and the full symmetry of the cyclic order is the dihedral group $\mathrm{O}_2(\mathbb{F}_7)$ of order 16.

There is no *total* order invariant under the rotations. An order-preserving permutation of a finite chain is the identity (the least element must be fixed, then inductively all the others), whereas the rotation by one step is an 8-cycle. One can choose a total order by cutting the circle at a base point and declaring $0 < 1 < \cdots < 7$, but the cut depends on the base point and is moved by the rotations; it is not canonical.

This is the finite analogue of the Euclidean circle. The circle has a *cyclic* order (an orientation), not a linear order; the real line has a linear order, the circle does not. The angle k is a coordinate on the cyclic order, not a linear order. The cyclic betweenness is also different from the linear one: for any three points, one lies on the oriented arc from one of the others to the third; this relation is preserved by rotations and reversed by reflections.

model	centre	affine stabiliser	angle group	cyclic order
ellipse	$(0, 0)$	$\mathrm{O}_2(\mathbb{F}_7) \cong D_8$	$\mathbb{Z}/8$	8 points
hyperbola	$(0, 0)$	$\mathrm{O}_{1,1}(\mathbb{F}_7) \cong D_6$	$\mathbb{Z}/6$	6 finite + 2 fixed
parabola	at infinity	$\mathrm{AGL}_1(\mathbb{F}_7)$	$\mathbb{Z}/7$	7 finite + 1 fixed

Table 4: The centre, the affine stabiliser, and the regular angle group for the three models. The centre is finite iff L_∞ is not tangent; the order of the regular angle group is $q + 1$, $q - 1$, q for the ellipse, hyperbola and parabola.

Table 4 records the three models. The regular angle group is the subgroup acting regularly on the finite points: for the ellipse and hyperbola it is the rotation part of the affine stabiliser, while

for the parabola it is the unipotent part. The parabola's full affine stabiliser is $\mathrm{AGL}_1(\mathbb{F}_7)$ of order 42, not the dihedral group D_7 of order 14; D_7 is only the subgroup with linear part $a = \pm 1$. The points fixed by the regular action are the intersections of L_∞ with C (none for the ellipse, two for the hyperbola, one for the parabola).

5.2 Arcs, gluing, and transitivity

A directed arc is an interval of the cyclic order. Write $[a, b]_\ell$ for the arc that starts at a and has length $\ell \in \{0, 1, \ldots, 8\}$; its point set is $\{a, a+1, \ldots, a+\ell\}$ modulo 8. The length ℓ is the angle difference $b - a$; $\ell = 0$ is the point a, and $\ell = 8$ is the full circle.

Gluing two arcs is defined when the endpoint of the first is the start of the second:

$$[a, b]_\ell \cdot [b, c]_m = [a, c]_{\ell+m} \qquad (\ell + m \leq 8).$$

If $\ell + m > 8$ the composite wraps more than once and is no longer a simple arc. Thus the simple arcs form a *partial* monoid, not a monoid. If one keeps the winding, i.e. lets the length be an integer in $\mathbb{Z}$ and records the projection $b - a \equiv \ell \pmod 8$, the arcs form the action groupoid of $\mathbb{Z}$ on $\mathbb{Z}/8$: objects are the eight points, arrows are the based arcs, composition is gluing, and the automorphism group of a point is $\mathbb{Z}$ (the full turns). If only nonnegative lengths are allowed, one gets the action category of $\mathbb{N}$; the endomorphism monoid of a point is $\mathbb{N}$ (full turns), not a group.

The question of transitivity has three different answers.

1. *The one-step relation.* For a nonempty set $S \subseteq \mathbb{Z}/8$ of allowed lengths, let $a \, R_S \, b$ iff $b - a \in S$. Then R_S is transitive if and only if S is a subgroup of $\mathbb{Z}/8$. The subgroups are

   $$\{0\}, \qquad \{0, 4\}, \qquad \{0, 2, 4, 6\}, \qquad \mathbb{Z}/8,$$

   with $8, 4, 2, 1$ classes respectively: equality, antipodal pairs, parity, and the universal relation.

2. *Reachability.* The relation R_S^* (a finite sequence of allowed steps) is transitive by construction. Its classes are the cosets of the subgroup $\langle S \rangle$ generated by S; the number of components is

   $$[\mathbb{Z}/8 : \langle S \rangle].$$

   The reachability graph has one component exactly when $\langle S \rangle = \mathbb{Z}/8$; the reachability relation itself is always transitive.

3. *The arc category.* With all lengths allowed, the arc groupoid is transitive (connected): any point reaches any other point. With a restricted set S of lengths, one gets the subgroupoid of $\langle S \rangle$; it is transitive on each coset, not on the whole circle unless S generates $\mathbb{Z}/8$.

Figure 4 shows the first three steps. Table 5 records the components and the one-step transitivity for representative sets.

The figure and the table make the point precise. The full arc structure is transitive, but a restricted arc structure need not be: the arcs of step 2 never connect an even point to an odd one, and the arcs of step 4 only connect antipodal pairs. The components are always the cosets of $\langle S \rangle$.

This is the finite analogue of the fundamental groupoid of the circle. The topological circle has path groupoid with automorphism group $\pi_1(S^1) = \mathbb{Z}$; the finite arc groupoid with winding has the same automorphism group $\mathbb{Z}$, and its projection to the eight points is the quotient by the full turn 8. The simple arcs form a partial composition structure (composition is defined only when the total length is at most 8); the action groupoid of $\mathbb{Z}$ (integer winding) is the groupoid with automorphism group $\mathbb{Z}$. Calling the simple arcs a "fundamental category" is only informal; the categorical structure with integer winding is the groupoid.

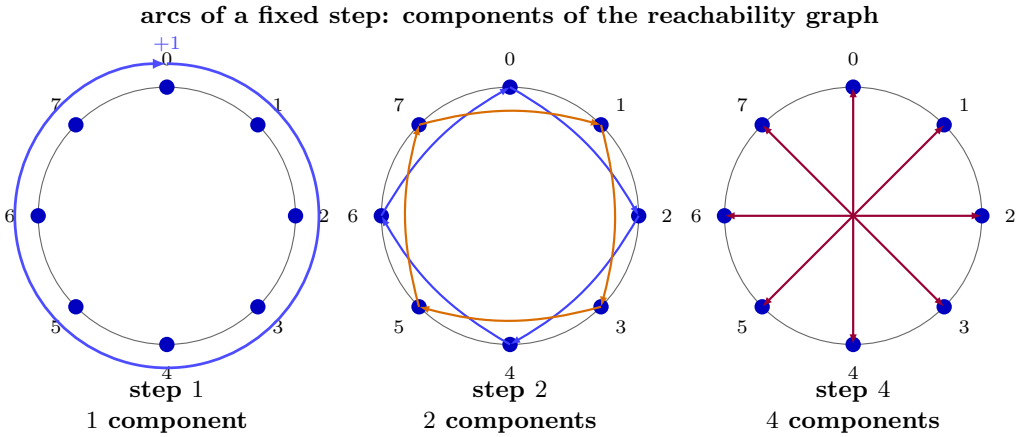


Figure 4: Arcs of a fixed step k: the components of the reachability graph. $k = 1$ gives one 8-cycle; $k = 2$ gives two 4-cycles (parity); $k = 4$ gives four 2-cycles (antipodal pairs).

S	$\langle S \rangle$	components	R_S transitive?
$\{1\}$	$\mathbb{Z}/8$	1	no
$\{2\}$	$\{0, 2, 4, 6\}$	2 (parity)	no
$\{3\}$	$\mathbb{Z}/8$	1	no
$\{4\}$	$\{0, 4\}$	4 (antipodal)	no
$\{2, 4\}$	$\{0, 2, 4, 6\}$	2	no
$\{0, 4\}$	$\{0, 4\}$	4	yes
$\mathbb{Z}/8$	$\mathbb{Z}/8$	1	yes

Table 5: Allowed arc lengths S, the subgroup they generate, the number of reachability components, and whether the one-step relation is transitive. For nonempty S the one-step relation is transitive exactly for the subgroups; reachability is always transitive, and it has exactly one component when S generates $\mathbb{Z}/8$.

5.3 Experiment: translating the circle

Translate every point of C by one cell upward:

$$T(x, y) = (x, y + 1).$$

The image of the conic is

$$C' : \quad x^2 + (y - 1)^2 = 1,$$

with the eight points

$$(1, 1),\ (2, 3),\ (0, 2),\ (5, 3),\ (6, 1),\ (5, 6),\ (0, 0),\ (2, 6).$$

The unit-circle relation is *not* preserved: none of the new points has norm 1 with respect to the original norm. Their norms are

$$2,\ 6,\ 4,\ 6,\ 2,\ 5,\ 0,\ 5,$$

so the new set is not the norm-one fibre of $N(x, y) = x^2 + y^2$. The centre moves from $O = (0, 0)$ to $O' = T(O) = (0, 1)$. The two centres exchange roles: the old centre O lies on C' because $0 + (0 - 1)^2 = 1$, and the new centre O' lies on C because $0 + 1^2 = 1$. The two conics are disjoint: $C \cap C'$ would require

$$y^2 = (y - 1)^2 \implies 2y = 1 \implies y = 4, \qquad x^2 + 16 = x^2 + 2 = 1 \implies x^2 = 6,$$

and 6 is a nonsquare in $\mathbb{F}_7$.

The image is nevertheless *still a conic*. The reason is that T is an affine transformation, and affine transformations preserve conics. In fact T fixes L_∞ pointwise, so C' is again an ellipse; the angle group is the conjugate

$$T\,G\,T^{-1} \cong \mathbb{Z}/8,$$

which fixes O' and acts regularly on C'. The cyclic order and the arc structure are transported by T; transitivity is unchanged.

	original C	translated C'
equation	$x^2 + y^2 = 1$	$x^2 + (y - 1)^2 = 1$
centre	$(0, 0)$	$(0, 1)$
norm $N(x, y) = x^2 + y^2$	constant 1	$0, 2, 4, 5, 6$
L_∞	disjoint	disjoint
angle group	G	$TGT^{-1} \cong \mathbb{Z}/8$

Table 6: The unit circle and its translate. The conic property and the projective/affine structure survive; the centre and the norm-one relation do not.

The moral is the distinction between the projective and the affine data. A conic is preserved by every projectivity, in particular by every translation; the unit-circle data — the centre O, the norm N, and the angle group around O — are not. A uniform translation gives another conic (a circle with a new centre), not the same circle. If the eight points were moved by an arbitrary permutation instead of a translation, the image would generally not be a conic at all; the conic property is projective/affine-invariant, not permutation-invariant.

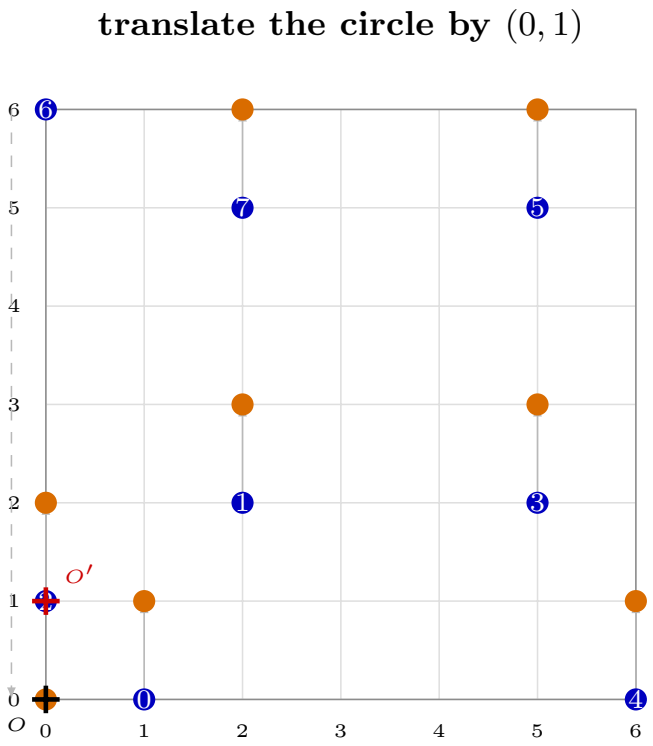


Figure 5: The original conic (blue, $x^2 + y^2 = 1$) and its translate (orange, $x^2 + (y-1)^2 = 1$) under $(x, y) \mapsto (x, y+1)$. The two conics are disjoint; the old centre $O = (0, 0)$ is a point of the orange conic, and the new centre $O' = (0, 1)$ is a point of the blue one.

5.4 Movable circles and transportability

The right definition is not a fixed subset but a relative relation. For a centre O and a radius parameter $r \in \mathbb{F}_7$, put

$$S(O, r) = \{P : Q(P - O) = r\}, \qquad Q(v) = v_1^2 + v_2^2.$$

The unit circle is $S(0, 1)$; the translated circle of the previous experiment is $S((0, 1), 1)$. The relation is translation-equivariant:

$$T\big(S(O, r)\big) = S\big(T(O), r\big)$$

for every translation T. The absolute norm $N(P) = Q(P)$ is not translation-invariant; the relative norm $Q(P - O)$ is. That is the movable circle relation. In Euclidean language r is the square of the radius; over $\mathbb{F}_7$ the square root need not exist, so r is the honest parameter.

The family of circles is

$$\mathcal{C} = \{(O, r) : O \in \mathbb{F}_7^2,\ r \in \mathbb{F}_7\}.$$

The translations act transitively on the centres with r fixed. For the linear part there are three different groups, and they must not be conflated:

- the orthogonal group $\mathrm{O}(Q)$: $Q(Av) = Q(v)$;
- the scalar similarities $\mathbb{F}_7^\times \mathrm{O}(Q)$: $A = \lambda R$, $Q(Av) = \lambda^2 Q(v)$;
- the full similarity group $\mathrm{GO}(Q)$: $Q(Av) = \mu Q(v)$ for some $\mu \in \mathbb{F}_7^\times$.

General quadratic-form theory uses the last group. Its elements need not be scalar multiples of orthogonal maps: for example

$$A = \begin{pmatrix} 1 & 4 \\ 3 & 1 \end{pmatrix}, \qquad A^T A = 3I,$$

so $Q(Av) = 3Q(v)$ and $A(S_1) = S_3$, although 3 is a nonsquare in $\mathbb{F}_7$.

group	definition	order	orbits on nonzero S_r
$\mathrm{O}(Q)$	$Q(Av) = Q(v)$	16	six
$\mathbb{F}_7^\times \mathrm{O}(Q)$	$A = \lambda R$	48	two square classes
$\mathrm{GO}(Q)$	$Q(Av) = \mu Q(v)$	96	one

Table 7: The three linear groups. $\mathrm{GO}(Q)$ is the full similarity group of the quadratic form; the scalar similarities are the subgroup with μ a square. The orbit counts on the six nonzero norm circles are 6, 2, and 1.

The circular points make the distinction precise. Homogenize $Q(x, y) = x^2 + y^2$ and put $z = 0$: the two points

$$I = [1 : \mathrm{i} : 0], \qquad J = [1 : -\mathrm{i} : 0]$$

in $\mathbb{P}^2(\mathbb{F}_{49})$ satisfy $x^2 + y^2 = 0$. A conic is a circle exactly when it passes through I and J. The stabilizer of the pair $\{I, J\}$ in the projective group is the full affine similarity group $\mathrm{GO}(Q)$; a general affine map acts on the line at infinity by its linear part and moves I, J unless the linear part lies in $\mathrm{GO}(Q)$, so it sends a circle to an ellipse; a general projectivity sends a circle to a conic not through I, J. The conic is projective; the circle is a similarity notion.

group	action on (O, r)	preserves circles?
translations	$O \mapsto O + t$, r fixed	yes
$\mathbb{F}_7^\times \mathrm{O}(Q)$	$O \mapsto \lambda RO + t$, $r \mapsto \lambda^2 r$	yes
$\mathrm{GO}(Q)$	$O \mapsto AO + t$, $r \mapsto \mu r$	yes
affine	$O \mapsto AO + t$	iff $A \in \mathrm{GO}(Q)$; otherwise ellipses
projective	general	iff $\{I, J\}$ is preserved; otherwise conics

Table 8: The circle is a similarity structure, not merely an affine or projective one. The full similarity group $\mathrm{GO}(Q)$ acts transitively on the nonzero circles; the scalar similarities $\mathbb{F}_7^\times \mathrm{O}(Q)$ have the two square-class orbits. I, J are the circular points at infinity.

The angle group is transported with the centre and the circle. The stabilizer of O *and* a nonzero circle S_r in the affine similarity group is $\mathrm{O}(Q)$; its rotation subgroup is $G = \mathrm{SO}(Q) \cong \mathbb{Z}/8$. For the translated circle the angle group is TGT^{-1}, and the cyclic order and the arc structure are transported by T. The stabilizer of O alone is the larger group $\mathrm{GO}(Q)$: it also contains the dilations that move the circle, and those are not angle rotations. Thus the whole rigid structure — centre, a chosen circle, angle group, cyclic order, arcs — is transportable under translations, and under the full similarity group up to the choice of the radius.

The square classes are the obstruction for the *scalar* similarities $\mathbb{F}_7^\times \mathrm{O}(Q)$, not for $\mathrm{GO}(Q)$:

$$\mathbb{F}_7^\times / \mathbb{F}_7^{\times 2} = \{\{1, 2, 4\}, \{3, 5, 6\}\}.$$

Under $\mathbb{F}_7^\times \mathrm{O}(Q)$ the unit circle is similar only to the square-radius circles. Under $\mathrm{GO}(Q)$ it is similar to every nonzero circle, because the multiplier μ may be a nonsquare. Over $\mathbb{R}$ the positive scalar similarities already act transitively on the positive radii; over $\mathbb{F}_7$ the scalar multipliers are squares, hence the two orbits, and $\mathrm{GO}(Q)$ adds the nonsquare multipliers that repair the gap.

5.5 Scaling and the ripple pencil

There are two different families of circles attached to the centre 0, and the scaling group acts on them differently.

The norm circles. The level sets

$$S_r = \{P : Q(P) = r\}, \qquad r \in \mathbb{F}_7^\times,$$

are the six 8-point conics centred at 0; $S_0 = \{0\}$. The dilation $D_\lambda(P) = \lambda P$ acts by

$$D_\lambda(S_r) = S_{\lambda^2 r}.$$

Since λ^2 runs over the squares $\{1, 2, 4\}$, the six circles split into two scaling orbits,

$$\{S_1, S_2, S_4\}, \qquad \{S_3, S_5, S_6\},$$

the square classes of the radius.

The ripple pencil. The affine clue horocycles in the finite model are the horizontal lines

$$H_t = \{x + ti : x \in \mathbb{F}_7\}, \qquad t \in \mathbb{F}_7,$$

with seven points each. In the projective line $\mathbb{P}^1(\mathbb{F}_{49}) = \mathbb{F}_{49} \cup \{\infty\}$ put $\widehat{H}_t = H_t \cup \{\infty\}$. The Möbius inversion

$$J(z) = -\frac{1}{z}$$

acts on $\mathbb{P}^1(\mathbb{F}_{49})$ and maps

$$J(\widehat{H}_t) = C_t : \quad x^2 + y^2 = \frac{y}{t}, \qquad t \in \mathbb{F}_7^\times.$$

Equivalently, on affine points, $J(H_t) = C_t \setminus \{0\}$: the missing point is exactly the pole $0 = J(\infty)$. Every C_t is an 8-point conic through 0, and all six share the tangent line $y = 0$ at 0. They form the finite *ripple pencil*. The correspondence

$$J(\widehat{H}_t) = C_t$$

is exact: the point path (the clue) and the ripple history are two readings of the same Möbius map, as in the path–ripple duality of Paper 0.

Three spaces must be kept apart: the affine plane $\mathbb{A}^2(\mathbb{F}_7)$ has 49 points; the projective plane $\mathbb{P}^2(\mathbb{F}_7)$ has 57; the Möbius line $\mathbb{P}^1(\mathbb{F}_{49})$ has 50. The infinity in $\widehat{H}_t$ is the one of $\mathbb{P}^1(\mathbb{F}_{49})$, not a line at infinity of $\mathbb{P}^2(\mathbb{F}_7)$.

Under the same dilation, the ripple parameter transforms by

$$D_\lambda(C_t) = C_{t/\lambda}.$$

Here λ runs over all of $\mathbb{F}_7^\times$, not just the squares, so the six ripple conics form a *single* orbit

$$\{C_1, C_2, \ldots, C_6\}.$$

This is the precise finite form of the nested family of Euclidean circles tangent at 0: the nesting parameter t is the height of the clue line, and the scaling acts on it by the inverse scalar λ^{-1}.

family	equation	common point	scaling	orbits
clue lines H_t	$y = t$	∞	$t \mapsto \lambda t$	one
ripple conics C_t	$x^2 + y^2 = y/t$	0	$t \mapsto \lambda^{-1} t$	one
norm circles S_r	$x^2 + y^2 = r$	none	$r \mapsto \lambda^2 r$	two square classes

Table 9: The clue lines H_t and the ripple conics $C_t = J(\widehat{H}_t)$ are exchanged by the inversion $J(z) = -1/z$; scaling acts by the inverse scalar on t. The orbit counts are for nonzero t and r; $t = 0$ (the x-axis) and $r = 0$ (the origin) are fixed. The norm circles S_r are a different family, with the square-class obstruction for scalar similarities.

The visual point is worth stating plainly. The Euclidean picture of the ripple family is a nested family of circles tangent at 0; the nesting is an ordered notion (inside/outside, radius). Over $\mathbb{F}_7$ there is no such order: the six C_t are 8-point conics through 0, and their arrangement on the 7×7 grid is a coloured partition, not a sequence of nested rings. What is observable is the *pencil* (all six through the pole), the *partition* of the grid, and the scaling action; what is not observable is the Euclidean nesting. The correspondence $J(\widehat{H}_t) = C_t$ and the scaling laws $t \mapsto \lambda^{-1} t$, $r \mapsto \lambda^2 r$ hold exactly; they are algebraic statements, not visual ones.

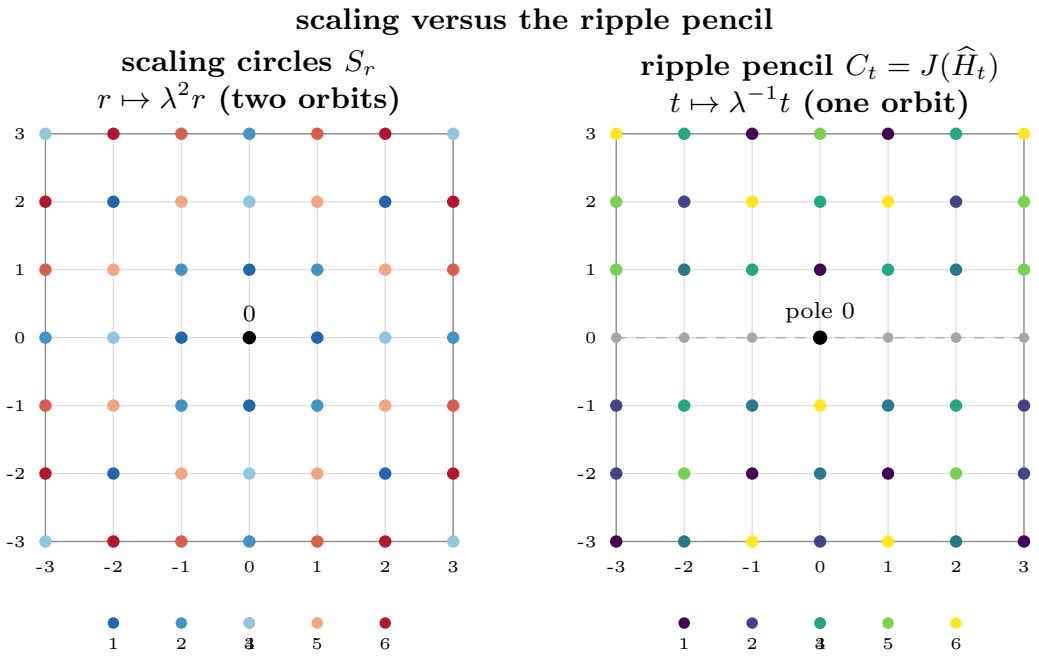


Figure 6: Left: the norm circles S_r (level sets of Q), coloured by r; their scaling orbits are the two square classes. Right: the ripple conics $C_t = J(\widehat{H}_t)$, all through the pole 0 with common tangent $y = 0$; the grey points on the x-axis are the clue line H_0. In the Euclidean upper half-plane the C_t are nested circles tangent at 0; over $\mathbb{F}_7$ they form a pencil of six 8-point conics through 0.

5.6 The real picture and the finite picture

It is worth placing the finite ripple pencil next to the real one from Paper 0. Over $\mathbb{R}$, with $J(z) = -1/z$ and the affine line $H_t = \{x + ti : x \in \mathbb{R}\}$, $t > 0$, put $\widehat{H}_t = H_t \cup \{\infty\}$ in the Riemann sphere $\mathbb{P}^1(\mathbb{C})$. The image is the circle

$$C_t : \quad x^2 + \left(y - \frac{1}{2t}\right)^2 = \left(\frac{1}{2t}\right)^2, \qquad t > 0.$$

All these circles are tangent to the real axis at 0 and nested as t varies; the parameter t is continuous. The compactified clue lines and the circles are exchanged by J:

$$J(\widehat{H}_t) = C_t.$$

On affine points $J(H_t) = C_t \setminus \{0\}$; the missing point is the pole $0 = J(\infty)$. The dilation D_λ, $\lambda > 0$, sends t to t/λ; the family is a single orbit, and the nesting is visible.

Over $\mathbb{F}_7$ the same equations produce the six conics

$$C_t : \quad x^2 + y^2 = \frac{y}{t}, \qquad t \in \mathbb{F}_7^\times.$$

They all pass through the pole 0 with the common tangent $y = 0$, and they form a *pencil*, not a nested sequence. The correspondence $J(\widehat{H}_t) = C_t$ and the scaling law $t \mapsto t/\lambda$ still hold; the six form one orbit. What is lost is the order: there is no inside/outside, no continuous parameter, no nesting.

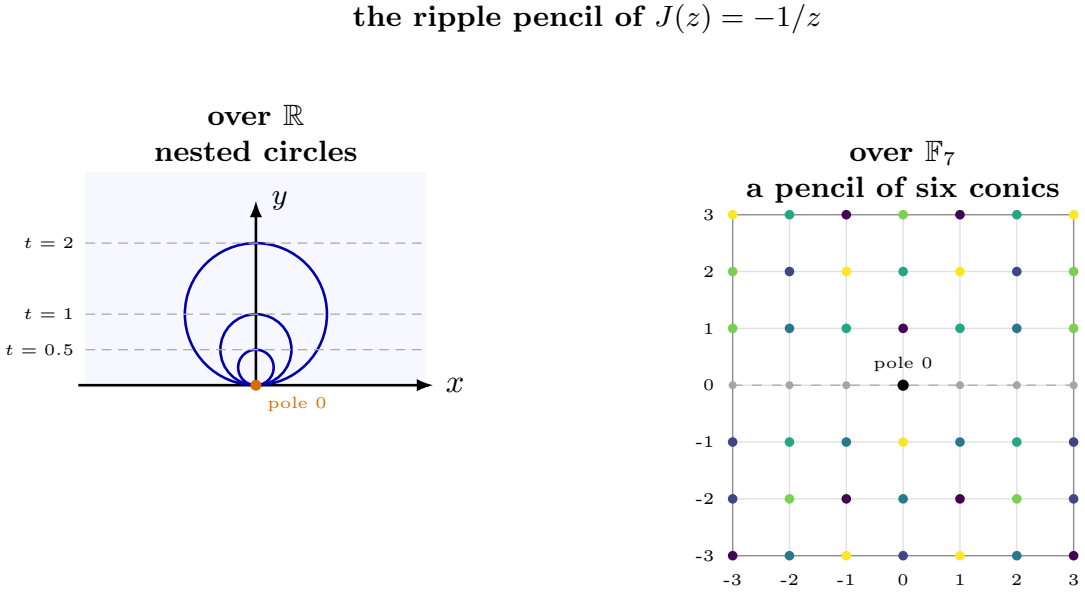


Figure 7: The ripple pencil of $J(z) = -1/z$ over $\mathbb{R}$ and over $\mathbb{F}_7$. Left: the compactified clue lines $\widehat{H}_t$ (dashed affine part H_t plus ∞) map to nested circles tangent at 0; the affine part misses the pole 0. Right: the same equations over $\mathbb{F}_7$ give six 8-point conics through 0, with common tangent $y = 0$; the family is a pencil, not a nested sequence.

The moral is the same one that has recurred throughout these notes: the *algebraic* structure — the conic family, the Möbius correspondence, the scaling orbit — is field-independent, while

	over $\mathbb{R}$	over $\mathbb{F}_7$
parameter	$t > 0$ continuous	$t \in \mathbb{F}_7^\times$ (six values)
C_t	nested circles tangent at 0	8-point conics through 0, common tangent $y = 0$
ordering	nesting by t	no order; pencil
scaling	$t \mapsto t/\lambda$, $\lambda > 0$	$t \mapsto t/\lambda$, $\lambda \in \mathbb{F}_7^\times$
orbits	one	one
$J(\widehat{H}_t) = C_t$	yes	yes
visual	nested rings	coloured pencil

Table 10: The real and finite ripple pencils. The algebraic correspondence and the scaling orbit are field-independent; the nesting is a real/order phenomenon.

the *order* structure — nesting, inside/outside, continuous parameter — is special to $\mathbb{R}$. The finite "ripple geometry" is a pencil, not a new geometry; the name is descriptive, as Paper 0 says of its real counterpart.

6 The angle group

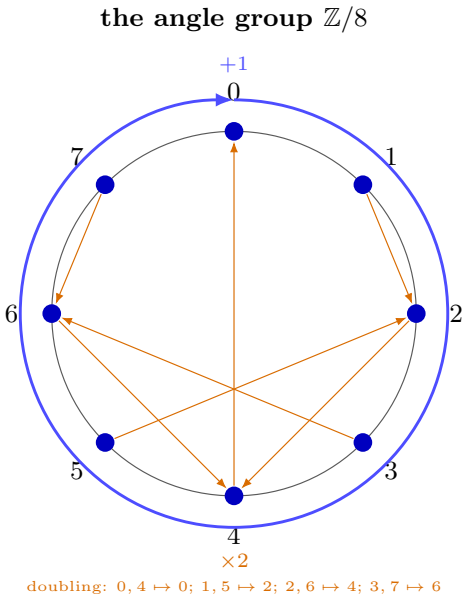


Figure 8: The angle group $\mathbb{Z}/8$. The blue outer arrow is the $+1$ cycle; the orange chords are the doubling map $k \mapsto 2k$. Doubling is two-to-one: $0, 4 \mapsto 0$; $1, 5 \mapsto 2$; $2, 6 \mapsto 4$; $3, 7 \mapsto 6$.

The angle group is the cyclic group $\mathbb{Z}/8$. Its automorphism group is $(\mathbb{Z}/8)^\times = \{1, 3, 5, 7\}$; multiplication by $3, 5, 7$ permutes the eight angles, while multiplication by 2 collapses pairs. The addition formula of Proposition 4.1 is the statement that the conic's group law and the angle group's addition are the same operation.

7 Euclidean angle axioms and the finite angle group

There are two different finite analogues of a Euclidean angle, and they must be kept apart:

- the *unit-vector angle* between two points of the conic $C : x^2 + y^2 = 1$, with values in $G = \mathbb{Z}/8$;
- the *line-direction angle* between two lines through the origin, with values in $D = \mathbb{F}_{49}^{\times}/\mathbb{F}_7^{\times}$, also isomorphic to $\mathbb{Z}/8$.

The first is a point-on-conic coordinate; the second is a direction ratio. The unit circle meets only four of the eight lines through the origin, each in a pair of antipodal points, so the unit-vector angle does not cover all directions. That is not a failure of line-direction angles; it is a statement about the unit-vector model. This section treats the unit-vector model first, then the line-direction model.

7.1 The Euclidean angle structure

The Euclidean angle group is $\mathbb{R}/2\pi\mathbb{Z}$ for oriented angles. It has:

1. *Congruence.* Angle congruence is an equivalence relation. Given an angle and a ray, there is a unique angle on a prescribed side of the ray congruent to the given one (Hilbert's copying axiom); SAS and the rest of Euclidean congruence follow.
2. *Protractor.* Angles have a real-valued additive measure, invariant under isometries.
3. *Cyclic order, not total order.* The circle has a cyclic order, preserved by rotations and reversed by reflections. There is no total order on $\mathbb{R}/2\pi\mathbb{Z}$ compatible with addition: the circle is not a chain. A total order on the universal cover $\mathbb{R}$ is a different structure; it is not an order on the angle group itself.
4. *Archimedean property and completeness.* These are separate. The Archimedean property rules out infinitesimal angles; completeness (the Dedekind/Cauchy property of $\mathbb{R}$) makes the angle line a continuum. Together they give the real-valued protractor.
5. *Divisibility.* For every $n \geq 1$ and every angle, $nh = k$ has a solution; this is the additive group of $\mathbb{R}$, not a consequence of order alone.

The standard theorems are: supplements and complements are unique; a right angle is congruent to its supplement; vertical angles are equal; every angle has a unique bisector; the angles of a triangle sum to π (equivalent to the parallel postulate in Euclidean geometry); the exterior-angle theorem; the isosceles base-angle theorem; the inscribed-angle theorem; Thales' theorem; and the AA criterion for similarity.

The finite affine plane also satisfies the parallel postulate: through a point not on a line there is a unique parallel. The failure of a Euclidean-style triangle angle sum in the unit-vector model below is therefore *not* a failure of the parallel postulate; it is a failure of having a Euclidean angle measure at arbitrary triangle vertices in that model.

In group language, oriented angles form $\mathbb{R}/2\pi\mathbb{Z}$; the rotation group $\mathrm{SO}(2)$ acts regularly on the circle, and the angle from a base ray to a target ray is the group element taking one to the other.

7.2 The unit-vector model

On the conic $C \cong \mathbb{P}^1(\mathbb{F}_7)$, the norm-one group $G = \mathbb{Z}/8$ acts regularly: for $P, Q \in C$ there is a unique $h \in \mathbb{Z}/8$ with $Q = \alpha^h P$. This h is the finite unit-vector angle from P to Q, and the map $h \mapsto \alpha^h$ is the finite protractor. Addition of angles is addition in $\mathbb{Z}/8$; the right angle is 2 (because $4 \cdot 2 = 8 = 0$), the straight angle is 4, and the full angle is 0. Figure 9 shows the two protractors side by side.

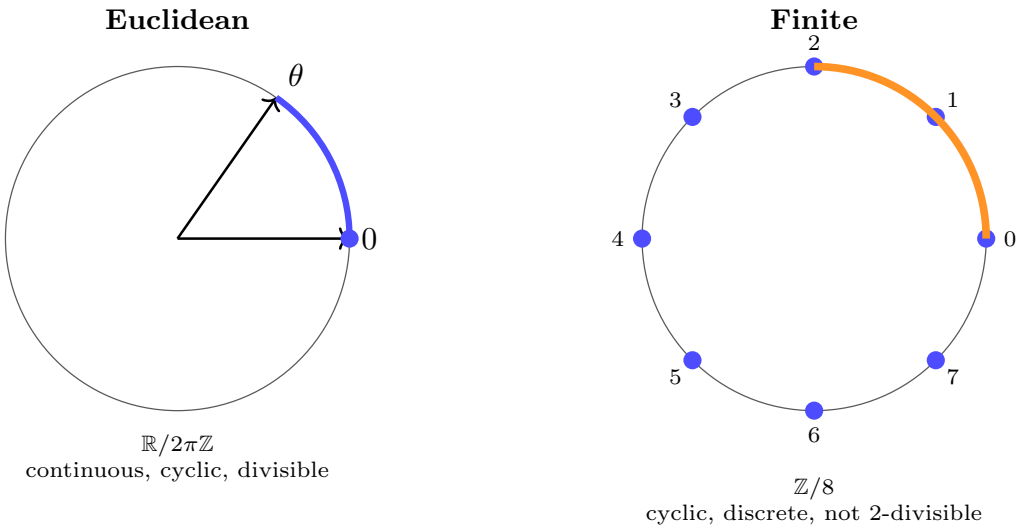


Figure 9: The continuous protractor $\mathbb{R}/2\pi\mathbb{Z}$ and the finite protractor $\mathbb{Z}/8$. On the right the eight points of C are the eight angles; the orange arc $0 \to 2$ is the right angle, passing through the angle 1.

Table 11 records the group-algebraic properties, and Table 12 the order, continuity and line-based ones.

Euclidean	finite unit-vector model	verdict
angle group $\mathbb{R}/2\pi\mathbb{Z}$	$\mathbb{Z}/8$ on C	different cyclic groups
protractor additivity	$k + l \bmod 8$	holds
right angle $\pi/2$; four right $= 2\pi$	2; $4 \cdot 2 = 8 = 0$	holds
supplement/complement	$k + l = 4$ / $k + l = 2$	holds
congruence is an equivalence	equality in $\mathbb{Z}/8$	holds
angle copying	translation $k \mapsto k + h$	holds
unique bisector	$2h = k$: 0 or 2 solutions	fails for odd k (e.g. $k = 1$)
n-section, all n	$nh = k$ iff $\gcd(n, 8) \mid k$	fails in general
Thales (semicircle)	$(A - P) \cdot (B - P) = 0$	holds for $P \neq A, B$
inscribed angle = half arc	$2h =$ arc, 2 not invertible	only 2-to-1 here

Table 11: Group-algebraic angle properties. "Holds" means a definition and proof; "fails" means a counterexample is given. The inscribed-angle row is the unit-vector reading; the line-direction reading is exact and appears below.

Euclidean	finite unit-vector model	verdict
cyclic order	cyclic order	both; shared
compatible total order	none	neither has one
Archimedean	no order/continuity	Euclidean only
completeness	no order/continuity	Euclidean only
continuous measure	$\mathbb{Z}/8$-valued	no continuous measure
vertical angles equal	no angle between lines	not defined in this model
triangle angle sum π	no angle at arbitrary vertices	not defined in this model; parallel postulate holds
similarity and AA	similarity group $\mathrm{GO}(Q)$ exists	AA not addressed

Table 12: Order, continuity and line-based properties. "Not defined in this model" is not a counterexample; it means the unit-vector angle has no such object. The finite affine plane satisfies the parallel postulate.

7.3 The line-direction model

The unit-vector model is not the only finite angle; the line-direction ratio is in the spirit of Wildberger's quadrance/spread [5]. Let

$$D = \mathbb{F}_{49}^{\times}/\mathbb{F}_7^{\times}.$$

Since $|\mathbb{F}_{49}^{\times}| = 48$ and $|\mathbb{F}_7^{\times}| = 6$, $|D| = 8$. The map

$$\rho : D \longrightarrow G, \qquad \rho([z]) = \frac{z}{\bar{z}},$$

is an isomorphism: its kernel is $\mathbb{F}_7^{\times}$ (the condition $z = \bar{z}$), and its image lies in the norm-one group G; both groups have order 8. A line through the origin has a direction in D. The eight points of C map to four lines (the antipodal pairs), while the other four directions have nonsquare nonzero norm and contain no $\mathbb{F}_7$-point of C. There are no nonzero isotropic vectors over $\mathbb{F}_7$: $x^2 + y^2 = 0$ would make -1 a square if $y \neq 0$, and if $y = 0$ it also forces $x = 0$. For a direction represented by v, a scalar multiple lies on C exactly when $Q(v)$ is a nonzero square, since $Q(cv) = c^2 Q(v)$.

For distinct $A, B, P \in C$,

$$\boxed{\rho\left(\frac{B-P}{A-P}\right) = \frac{B}{A}} \tag{1}$$

and the left side is independent of P. The proof uses $\bar{B} = 1/B$ and $\bar{P} = 1/P$:

$$\overline{B-P} = \bar{B} - \bar{P} = \frac{1}{B} - \frac{1}{P} = \frac{P-B}{BP} = -\frac{B-P}{BP},$$

so

$$\rho\left(\frac{B-P}{A-P}\right) = \frac{(B-P)/(A-P)}{\overline{(B-P)/(A-P)}} = \frac{B-P}{A-P} \cdot \frac{\overline{A-P}}{\overline{B-P}} = \frac{B-P}{A-P} \cdot \frac{-(A-P)/(AP)}{-(B-P)/(BP)} = \frac{BP}{AP} = \frac{B}{A}.$$

This is the exact line-direction inscribed-angle relation; it does not require a half-angle in $\mathbb{Z}/8$. It is checked for all 336 ordered triples in `verify.py`.

The line-direction angle between two lines with directions $d_1, d_2 \in D$ is $d_1/d_2 \in G$. Vertical angles are equal (the two pairs have the same ratio). For a triangle with side directions d_1, d_2, d_3, the product of the three turning ratios is 1, so the oriented turning sum is 0 (the full turn); this is the exterior-angle statement, not the interior-angle sum π. The interior sum is a different statement and is not implied by the group law.

	unit-vector angle	line-direction angle
values	$G = \mathbb{Z}/8$	$D \cong G$
domain	points of C	lines through 0
coverage	4 lines (8 points)	all 8 directions
inscribed angle	$2h =$ arc (2-to-1)	$\rho((B-P)/(A-P)) = B/A$
vertical angles	not defined	equal

Table 13: The two finite angles. The unit-vector angle is a point-on-conic coordinate; the line-direction angle is a direction ratio. Neither is a failure of the other.

7.4 Three concrete comparisons

Example 7.1 (Bisection)**.** In Euclidean geometry every angle $0 < \alpha < \pi$ has a unique bisector. In the unit-vector group $\mathbb{Z}/8$ the bisector equation is $2h = k$. If k is even there are two solutions h and $h+4$, the two orientations of the same bisector line; the finite group has no total order with which to choose one. If k is odd there is no solution. For example, the right angle $k = 2$ has $h = 1, 5$, while $k = 1$ has no bisector. This is the failure of divisibility by 2.

Proposition 7.2 (Finite Thales)**.** *Let $A, B \in C$ be antipodal, so $B = -A$, and let $P \in C$ with $P \neq A, B$. Then*

$$(A-P)\cdot(B-P) = A\cdot B - A\cdot P - P\cdot B + P\cdot P = -1 - A\cdot P + A\cdot P + 1 = 0.$$

Thus the vectors PA and PB are orthogonal with respect to the standard dot product on $\mathbb{F}_7^2$: the angle in a semicircle is right, in the bilinear sense. The algebraic identity also holds for $P = A$ or $P = B$, but then one of the vectors PA, PB is zero, so it is not an angle statement.

Proof. $A\cdot B = -1$ because $B = -A$ and $A\cdot A = 1$; $P\cdot P = 1$ because $P \in C$; and $P\cdot B = P\cdot(-A) = -A\cdot P$. The sum is zero. □

Remark 7.3 (Triangle angle sum and inscribed angle). The unit-vector model has no angle at an arbitrary vertex of a finite triangle: the group $\mathbb{Z}/8$ is attached to the conic, not to arbitrary lines. The line-direction model does define the turning angles at a triangle; their oriented sum is 0 (the full turn), not the Euclidean interior sum π. The finite affine plane satisfies the parallel postulate, so the Euclidean triangle angle sum is not a counterexample to it; the two statements are about different angle structures. Likewise, the unit-vector half-arc equation $2h =$ arc is only 2-to-1, while the line-direction relation (1) is exact and requires no halving.

7.5 What the comparison says

The common algebraic skeleton is the angle group: a cyclic group acting regularly on the circle or conic, with the angle as the group element. Euclidean geometry adds an Archimedean continuous measure, completeness, and divisibility. It does *not* add a compatible total order: neither the Euclidean circle nor the finite one has one; both have a cyclic order. The finite affine plane also satisfies the parallel postulate. The difference is therefore continuity and divisibility, not total order and not the existence of parallels.

The theorems that depend only on the group law survive: addition, supplements, complements, right and straight angles, copying, and finite Thales. The theorems that need order, continuity, or division fail or are not defined: bisection, n-section, the unit-vector triangle angle sum, and continuous measure. The line-direction model supplies an exact angle for arbitrary lines and the

exact inscribed relation (1); the unit-vector model supplies the point-on-conic coordinate. This is the same pattern as the $\mathbb{R}/\mathbb{F}_7$ trichotomy: the algebraic classification survives; the continuous invariants do not.

8 Stereographic and half-angle coordinates

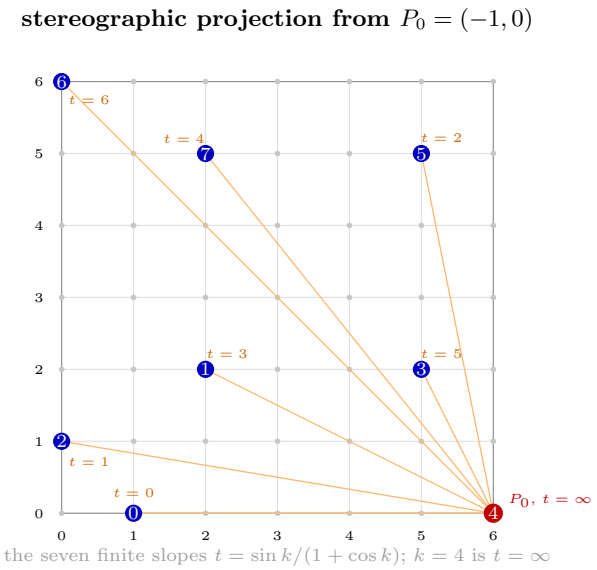


Figure 10: Stereographic projection from $P_0 = (-1, 0) = (6, 0)$. The seven finite slopes $t = \sin k/(1 + \cos k)$ give the seven points other than P_0; the point P_0 itself is $t = \infty$. The Euclidean segments are schematic: a projective line over $\mathbb{F}_7$ contains eight points and wraps around the grid.

For $k \neq 4$ (so $\alpha^k \neq -1$), the line through $P_0 = (-1, 0)$ and $(\cos k, \sin k)$ has slope

$$t = \frac{\sin k}{1 + \cos k} \in \mathbb{F}_7. \tag{2}$$

The inverse formulas are

$$\cos k = \frac{1 - t^2}{1 + t^2}, \qquad \sin k = \frac{2t}{1 + t^2}, \tag{3}$$

and $k = 4$ corresponds to $t = \infty$. Thus

$$k \longmapsto t$$

is a bijection $\mathbb{Z}/8 \setminus \{4\} \to \mathbb{F}_7$, i.e. a finite half-angle (tangent) coordinate. The formulas (2)–(3) are verified for all k in `verify.py`.

9 Discrete waveforms and the first difference

The functions cos and sin are periodic functions on the finite group $\mathbb{Z}/8$. Their first finite difference is

$$\Delta f(k) = f(k + 1) - f(k).$$

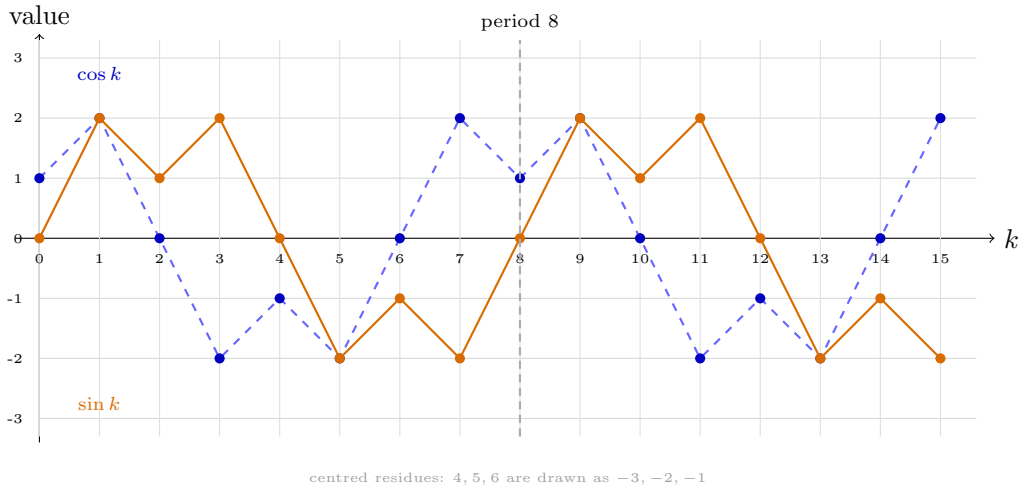


Figure 11: $\cos k$ and $\sin k$ as functions on $\mathbb{Z}/8$, repeated over two periods. The values are centred residues in $\{-3, \ldots, 3\}$; the dashed vertical line marks the period 8. These are discrete waveforms, not continuous graphs.

For example, with centred representatives,

$$\Delta \cos = (1, -2, -2, 1, -1, 2, 2, -1), \qquad \Delta \sin = (2, -1, 1, -2, -2, 1, -1, 2),$$

and in both cases the sum over the cycle is 0:

$$\sum_{k=0}^{7} \Delta f(k) = f(8) - f(0) = 0.$$

This is the discrete fundamental theorem of calculus: summation and difference are inverse up to the value at 0. Conversely, for $g : \mathbb{Z}/8 \to \mathbb{F}_7$, the equation $\Delta f = g$ has a solution if and only if

$$\sum_{k=0}^{7} g(k) = 0.$$

The solutions are then $f + c$ for a constant c: the kernel of Δ is the constants, and its image is the sum-zero functions. This is the periodic Fredholm alternative; it is the obstruction to a periodic antiderivative.

There is no limit, no derivative as a limit, and no mean value theorem; the calculus here is finite and algebraic. A caution about function spaces: a polynomial *function* on $\mathbb{F}_7$ is not the same object as a polynomial. The polynomial $x^7 - x$ vanishes at every point of $\mathbb{F}_7$, but its formal derivative $7x^6 - 1 = -1$ is not zero. The derivation lives on polynomials and rational functions, not on the algebra of $\mathbb{F}_7$-valued functions; to descend it to point functions one must quotient by $x^7 - x$, and the derivative does not respect that quotient. This is why the discrete calculus here is formal and algebraic, not a calculus of arbitrary point functions.

The next section gives the finite-field Fourier layer. A systematic treatment of Δ, summation, Hasse derivatives and residues on $\mathbb{P}^1(\mathbb{F}_7)$ is the next note.

10 The harmonic layers

The algebraic trigonometry of the previous sections takes values in $\mathbb{F}_7$. There are two harmonic layers above it, and they must not be conflated: a finite-field layer with values in $\mathbb{F}_{49}$, and a complex layer with values in $\mathbb{C}$. The first is the algebraic bridge; the second is a unitary representation.

10.1 The finite-field layer

Let

$$\chi_j(k) = \alpha^{jk} \in \mathbb{F}_{49}, \qquad j, k \in \mathbb{Z}/8.$$

These are the finite-field characters of $\mathbb{Z}/8$. Orthogonality reads

$$\sum_{k=0}^{7} \chi_j(k)\chi_{-l}(k) = 8\,\delta_{jl}.$$

The field $\mathbb{F}_{49}$ has characteristic 7, so $8 = 1 \neq 0$; the sum is 1 for $j = l$ and 0 otherwise. For a function $f : \mathbb{Z}/8 \to \mathbb{F}_{49}$ define

$$\widehat{f}(j) = \sum_{k=0}^{7} f(k)\chi_{-j}(k), \qquad f(k) = 8^{-1} \sum_{j=0}^{7} \widehat{f}(j)\chi_j(k),$$

with $8^{-1} = 1$. The Fourier matrix $(\chi_j(k))_{j,k}$ is invertible; the transform is an isomorphism of the 8-dimensional space of functions.

The difference operator acts on the characters by

$$\Delta\chi_j = (\alpha^j - 1)\chi_j.$$

Thus the characters are the eigenvectors of Δ, with eigenvalues $\alpha^j - 1$. The eigenvalues are distinct because α^j are distinct; the Fourier transform diagonalizes Δ. The constants are $j = 0$ with eigenvalue 0, and the image of Δ is the sum-zero functions, in agreement with the solvability condition of §9.

The trigonometric functions are the real and imaginary coordinates in this basis:

$$\cos k = \frac{\chi_1(k) + \chi_{-1}(k)}{2}, \qquad \sin k = \frac{\chi_1(k) - \chi_{-1}(k)}{2\mathrm{i}}.$$

Their Fourier transforms have support on $j = \pm 1$. In this sense the finite-field Fourier layer is the algebraic bridge between the cos / sin tables and the difference operator: the same α produces the point coordinates, the characters, and the spectrum.

10.2 The complex layer

A second harmonic layer takes values in $\mathbb{C}$. Let

$$\zeta_8 = e^{2\pi\mathrm{i}/8}$$

and define the complex characters of $\mathbb{Z}/8$ by

$$\chi_j(k) = \zeta_8^{jk}, \qquad j, k \in \mathbb{Z}/8.$$

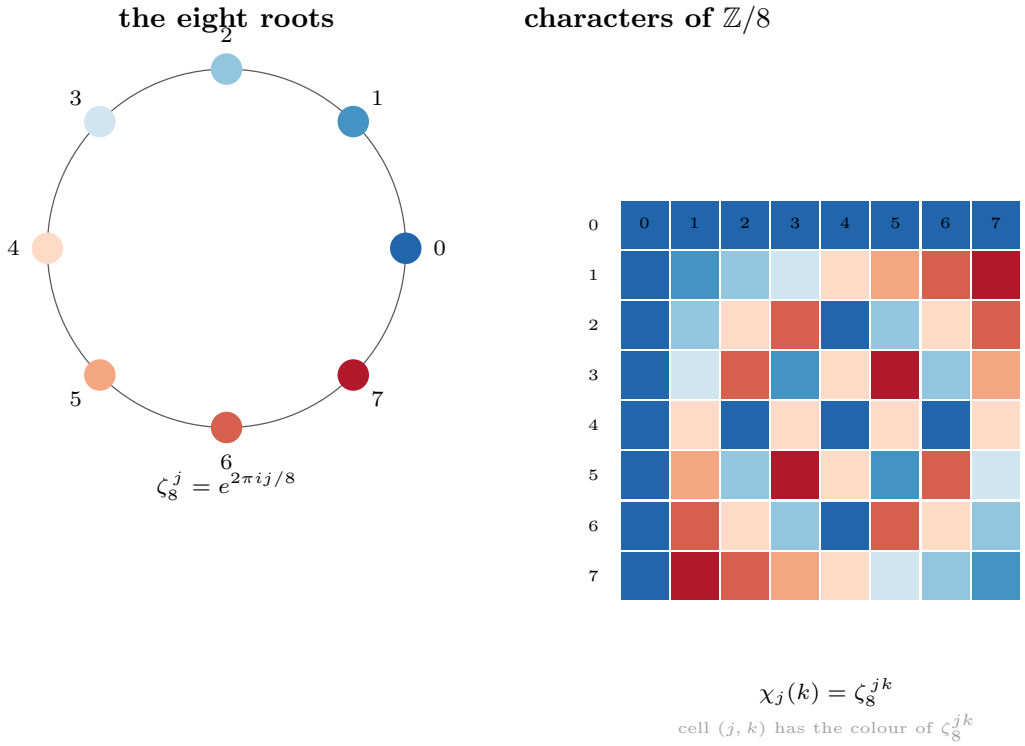


Figure 12: The complex harmonic layer. Left: the eight eighth roots of unity ζ_8^j. Right: the character matrix $\chi_j(k) = \zeta_8^{jk}$; the cell (j, k) has the colour of ζ_8^{jk}.

For a function $f : \mathbb{Z}/8 \to \mathbb{C}$, the finite Fourier transform is

$$\widehat{f}(j) = \sum_{k=0}^{7} f(k)\, \zeta_8^{-jk},$$

with inversion

$$f(k) = \frac{1}{8} \sum_{j=0}^{7} \widehat{f}(j)\, \zeta_8^{jk}$$

and Parseval

$$\sum_{k=0}^{7} |f(k)|^2 = \frac{1}{8} \sum_{j=0}^{7} |\widehat{f}(j)|^2.$$

The transform diagonalizes translations: if $(\tau_a f)(k) = f(k-a)$, then $\widehat{\tau_a f}(j) = \zeta_8^{-ja} \widehat{f}(j)$. For the general theory of Fourier analysis on finite groups see [2].

10.3 The two layers

The finite-field characters $\chi_j^{\mathbb{F}}(k) = \alpha^{jk} \in \mathbb{F}_{49}$ and the complex characters $\chi_j^{\mathbb{C}}(k) = \zeta_8^{jk} \in \mathbb{C}$ share the index group $\mathbb{Z}/8$ and the same orthogonality pattern, but they take values in different fields. The finite-field layer is algebraic: it diagonalizes Δ, and the cos / sin coordinates are its ± 1 modes. The complex layer is unitary: it gives the Hilbert-space transform and Parseval. Neither is a substitute for the other.

The bridge between the additive characters of $\mathbb{F}_7$ and the multiplicative characters of $\mathbb{F}_7^\times$ is the Gauss sum

$$G(\chi) = \sum_{x \in \mathbb{F}_7} \chi(x)\, \zeta_7^x,$$

which is the finite analogue of the Gamma integral; it will be treated in the next note. A further bridge between finite point functions, rational functions on $\mathbb{P}^1(\mathbb{F}_7)$, and functions on local fields is the subject of the discrete-calculus note; the polynomial $x^7 - x$ of §9 is the basic warning that these are different function spaces.

11 What is not claimed, and what comes next

This note is deliberately small. It does *not* claim:

- a real or complex analysis over $\mathbb{F}_7$: there are no nontrivial limits, no continuity in the analytic sense, no mean value theorem;
- that the finite trigonometry is a picture of the real one: the values are in $\mathbb{F}_7$, the angle group is $\mathbb{Z}/8$, and the identities are algebraic;
- that the conic or Pascal's theorem alone gives analysis: the conic gives the geometry and the group law; the analysis comes from the derivation, difference, representation and cohomological structures built on top;
- that the finite-field Fourier layer and the complex Fourier layer are the same representation: they share the index group but take values in different fields.

The finite-field Fourier layer of Section 10 is now included. The next installment, *Conic Motions, Arithmetic Histories, and Three Finite Calculi* (`../conic-dynamics/conic-dynamics.tex`), now gives the arithmetic-history realization of the eight-point circle step, its ordinary evaluation domains, and the comparison with split six-cycles and unipotent seven-cycles. In characteristic seven the last has a nilpotent difference, whereas the first two admit Fourier diagonalization over appropriate fields. Further directions after this history/calculus interface are:

1. discrete calculus on $\mathbb{P}^1(\mathbb{F}_7)$: Δ, summation, Hasse derivatives, differential forms, residues and the residue theorem, together with the function-space distinctions (the polynomial $x^7 - x$ versus the function it defines);

2. Gauss sums and the bridge between the additive and multiplicative characters;

3. the harmonic analysis of functions on $\mathrm{PGL}_2(\mathbb{F}_7)$ and the Bruhat–Tits tree [1], as a non-Archimedean analogue of the hyperbolic analysis of Paper [6]/Paper [7];

4. the characteristic-7 Magnus expansion and the group algebra $\mathbb{F}_7[\mathrm{PGL}_2(\mathbb{F}_7)]$, as the finite analogue of the series pillar of Paper [7].

Scope. Sections 1–4 and 7–10 form the trigonometry and difference core; Sections 5–6 (transformations, centres, arcs, ripple) are a geometric excursion. They can be split into a second paper once the foundations settle; the present revision keeps them together so that the cross-references remain explicit.

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